Lab Final

CSE 4308

DATABASE MANAGEMENT SYSTEMS LAB

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1 Lab Final

1. Rank films using a weighted value calculation: (rental_rate $\times$ 0.5) + (length $\times$ 0.1) + (replacement_cost $\times$ 0.2). Based on the value, list only the top 3 films with their title, release year, and calculated score.

2. Find the top 5 customers who have spent the highest total amount. Show customer ID, first name, last name, and total amount spent.

3. List all film titles consisting of more than one word that have been rented by customers from at least two different cities. Also list the total number of unique customers who rented each of those films, sorted by the customer count in descending order.

4. List the country having the most number of cities without any registered customers.

5. Write a procedure named `update_customer_email` that accepts a customer ID and a new email string. Update the targeted customer's email address and refresh their `last_update` timestamp.

6. Write an anonymous block that checks whether a given customer ID currently has any active rentals. Print a notice stating either: (i) "Customer has active rentals", or (ii) "Customer has no active rentals".

7. Write an anonymous block to show all films with a replacement cost greater than 28.00.

8. Write an anonymous block that defines a customer ID and a start date. Calculate and print their total spend from that start date up to the current date.

9. Create a function that prints customers who did not rent any film in the last N days, taking the number of days as an input parameter.

10. Calculate each customer's percentage of their store's total revenue. Show their name and store ID.

2 ER Diagram


actor
actor_id INT * pk
first_name VARCHAR(45) *
last_name VARCHAR(45) *
last_update DATETIME *
language
language_id INT * pk
name CHAR(20) *
last_update DATETIME *
film
film_id INT * pk
title VARCHAR(255) *
description TEXT
release_year YEAR
language_id INT *
rental_duration SMALLINT *
rental_rate DECIMAL(4,2) *
length SMALLINT
replacement_cost DECIMAL(5,2) *
rating film_rating_enum
last_update DATETIME *
special_features TEXT
fulltext_col TEXT *
category
category_id INT * pk
name VARCHAR(25) *
last_update DATETIME *
country
country_id INT * pk
country VARCHAR(50) *
last_update DATETIME *
city
city_id INT * pk
city VARCHAR(50) *
country_id INT *
last_update DATETIME *
address
address_id INT * pk
address VARCHAR(50) *
address2 VARCHAR(50)
district VARCHAR(20) *
city_id INT *
postal_code VARCHAR(10)
phone VARCHAR(20) *
last_update DATETIME *
film_actor
actor_id INT *
film_id INT *
last_update DATETIME *
inventory
inventory_id INT * pk
film_id INT *
store_id INT *
last_update DATETIME *
film_category
film_id INT *
category_id INT *
last_update DATETIME *
customer
customer_id INT * pk
store_id INT *
first_name VARCHAR(45) *
last_name VARCHAR(45) *
email VARCHAR(50)
address_id INT *
activebool BOOLEAN *
create_date DATE *
last_update DATETIME
active INT
staff
staff_id INT * pk
first_name VARCHAR(45) *
last_name VARCHAR(45) *
address_id INT *
email VARCHAR(50)
store_id INT *
active BOOLEAN *
username VARCHAR(16) *
password VARCHAR(40)
last_update DATETIME *
picture BLOB
rental
rental_id INT * pk
rental_date DATETIME *
inventory_id INT *
customer_id INT *
return_date DATETIME
staff_id INT *
last_update DATETIME *
store
store_id INT * pk
manager_staff_id INT *
address_id INT *
last_update DATETIME *
payment
payment_id INT * pk
customer_id INT *
staff_id INT *
rental_id INT
amount DECIMAL(5,2) *
payment_date DATETIME *